

Causal Regularity Theory III: Casebook of Causal Regularities and Benchmarks

The Role of Intuition, Philosophical Sprachspiele, Qualitative and Experimental Cases, and Design Survey

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Abstract

A regularity theory of causation is tested by whether its definitions recover — and, where the intuition is confused by co-occurrence, *correct* — the pre-theoretic verdicts of a fixed case repertoire. This volume is that repertoire. It is organised so that each fully worked case exhibits *which condition of the theory does the work*: minimal sufficiency against idle conjuncts, minimal necessity against the Manchester hooters, bundle-level homogeneity against naive correlation, Gate B against a confounded risk difference. Twenty-four philosophical *Sprachspiele*, the qualitative coincidence analyses of Volume I, and the experimental inferences of Volume II are one casebook, not three. Stub summaries are refused: a case is either carried through the inference panel or recorded in the master table with its deciding rule.

keywords causal casebook · causal intuitions · benchmark cases · minimal theories · Sprachspiele · coincidence analysis · experimental design survey · causal hypothesis inference · fire causation · statin trial · Simpsons paradox · LK-99 · OPERA

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i Note

Causal Regularity Bundle — Volume III. Companion volumes: I (qualitative theory), II (experimental inference). Formal calculus: `models.qmd`. Visual language: `causal-visuals.qmd`. Terms are used in the senses of the LWgrammar register *causal-regularity*.

1 The role of intuition in causal theory

💡 The claim of this volume

A theory of causation earns its keep by recovering the stable verdicts of a fixed case repertoire from its definitions, and by correcting the unstable ones. Intuition is the test, not the analysans.

1.1 Causal intuitions as Sprachspiele

Theories of causation are tested against short case descriptions. The method is uniform across the literature: a scenario is told in a few sentences; a verdict about the relevance of one factor to another is elicited; the theory under examination is required to replicate that verdict from its definitions alone. The theory that meets this test speaks of a few things only, and a reader who has not come from Volume I needs them before the first definition:

Note

Key. A *factor* is a kind of condition or event that is either present (value 1) or absent (value 0) in a case — “the match is struck”, “the Londoners go home”. A *literal* is a factor with its sign: present (A) or absent ($\neg A$). A *cluster* is a combination of literals that is enough for the outcome and contains nothing idle (in the formulae: a minimally sufficient condition, MSC). A *minimal theory* of an outcome lists the clusters that are needed for it, none of them spare. The *factor set* $\mathcal{F}$ is the explicit list of factors a case records; the *causal field* is the range of cases a claim is about. *Minimal sufficiency* is the test that no conjunct of a cluster can be dropped; *minimal necessity* the test that no cluster can be dropped from the theory; *homogeneity* the requirement that no unrecorded rival cluster acts differently across the compared cases; *instantiation* asks which cluster actually fired in a particular case.

In formulae $*$ reads *and*, $+$ reads *or*, $\neg$ reads *not*. $X \rightarrow E$ reads “ X is sufficient for E ”; $X \leftrightarrow E$ “ X and E are present in exactly the same cases”; $X \Rightarrow E$ “ X is a minimal theory of E , written in the causal direction”. Lower-case letters (a , e) name particular occurrences — the *tokens* of the factor types A , E . The definitions themselves are those of Volume I.

Definition — causal Sprachspiel

A **causal Sprachspiel** is a short case description in plain language for which competent speakers converge on a causal verdict before consulting any theory. The theory must then derive that verdict from its own conditions — minimal sufficiency, minimal necessity, homogeneity, instantiation — or state explicitly why it dissents.

Stated exactly. A causal Sprachspiel is a triple $(D, V, \mathcal{F})$ where D is a case description, V is the inter-subjective verdict (“ A caused E ” / “ A did not cause E ”), and $\mathcal{F}$ is the factor set the description makes available. The regularity theory is tested by whether it recovers V from $(D, \mathcal{F})$ using only the named conditions.

Example. The Manchester hooters (Mackie 1974, 81–84): speakers judge that the factory hooters do not cause the Londoners to go home; Mackie’s INUS analysis (a cause is an insufficient but non-redundant part of an unnecessary but sufficient condition) certifies them as a cause; minimal necessity prunes them (Volume I, section “The Manchester hooters: what minimal necessity does”; this volume, Section 2.5). The theory earns the verdict by recovering what plain-language use already secured.

The term is used here in Wittgenstein’s sense: “the meaning of a word is its use in the language” (PU §43), and grammar “tells what kind of object anything is” (PU §373). The casebook records use, not a hidden essence of *cause*.

This volume assembles the repertoire of the modern regularity-theoretic tradition: Graßhoff and May (2001), Baumgartner (2008, 2013), Baumgartner and Falk (2023), and Andreas and Günther (2019, 2024a, 2024b). Each fully worked case records three items:

1. **Situation** — the description, quoted from the source where possible.
2. **Causal intuition** — the pre-theoretic verdict, as reported in the source.
3. **Regularity judgement** — the verdict the theory derives, with the condition that secures it — or an account of dissent.

A factor A is causally relevant to E if and only if A (or its negation) is a non-redundant conjunct of a cluster that is itself a non-redundant disjunct of a minimally necessary condition of E — a *minimal theory* of E (Graßhoff and May 2001) — and this membership is stable under suitable expansions of the factor set (Baumgartner 2013; Baumgartner and Falk 2023). The casebook shows, case by case, which of these conditions does the work.

1.2 Replication, correction, explication

Formal regularity theory stands to intuition in three relations, and only three:

- **Replication.** Where the intuition is stable, the theory must recover it from the coincidence table, without unobserved primitives.
- **Correction.** Where co-occurrence misleads (the hooters; Simpson’s reversal), the theory exposes the structural flaw — failure of minimal necessity (the hooters), or of homogeneity (Gate B, Simpson’s reversal).
- **Explication.** Pre-theoretic language conflates type-level with token-level causation, sufficiency with necessity, risk with protective polarity. The theory distinguishes them.

1.3 The inference panel

Every fully worked case in Parts 3 and 4 follows the six-step inference procedure of Volume II (causal hypothesis inference):

1. state the causal hypothesis;
2. secure Gate B;
3. run Gate A;
4. estimate the unique contribution of the treatment’s cluster;
5. recover and score candidate clusters;
6. state the licensed conclusion.

The panels set these steps out under the headings query, causal field and factor set, data setup, gates, recovered hypothesis with its scores, and what would change the verdict.

Note

Key. A *contradictory configuration* is a combination of recorded factor values that occurs with the outcome present and with it absent. C_E is the number of such configurations for outcome E ; B_E , the *contradiction burden*, is the smallest share of rows that would have to be removed or reclassified

to leave none. **Gate A** asks whether a table is clean enough to read a complete theory from it: no contradictory configuration, $C_E = 0$. **Gate B** asks whether a two-arm comparison isolates the treatment factor: no rival cluster fires differentially with it (secured by randomisation, blocking, or an explicit bound). $\hat{\Delta} = p_1 - p_0$ is the risk difference between the outcome rates with and without the treatment; *con* (consistency) is the share of cases with the cluster that have the outcome, *cov* (coverage) the share of outcome cases that have the cluster. INF-01–11 are the inference warrants and FALL-01–10 the fallacy inhibitors of the standard registry (Volume I).

The philosophical cases (Section 2 to Section 6) use the staged panel of `models.qmd`: normalisation, coincidence table, derivation, verdict — and a CVL graph, the visual language specified in `causal-visuals.qmd`.

One convention carries every diagram in this volume, and a reader who has not come from Volume I needs only this much of it:

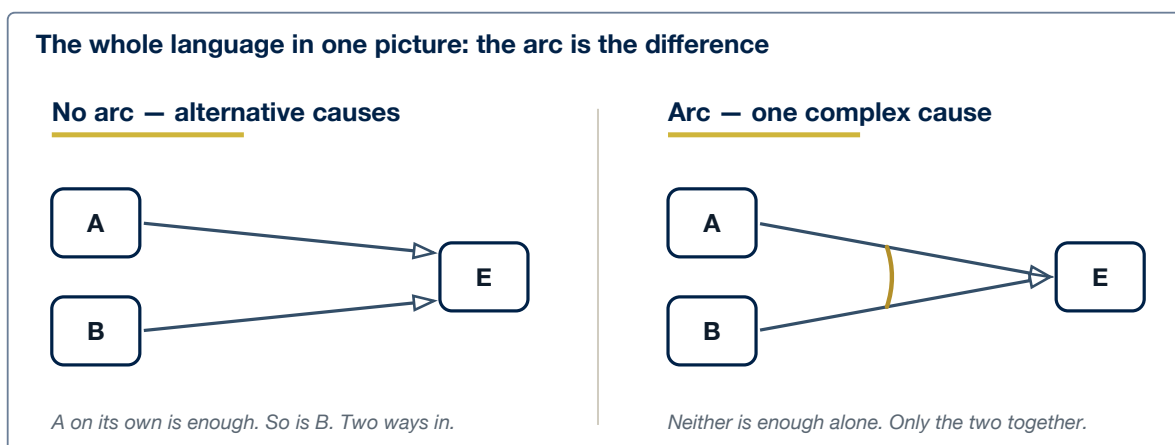


Figure 1 How to read every diagram in the casebook. Arrows run from causes to the effect. Arrows tied by an arc are the jointly required parts of *one* complex cause — remove any of them and the cause is not there. Arrows without an arc are *alternative* causes, each sufficient on its own. Two further marks appear in the cases. A small box marked $\neg$ on the shaft, where a line leaves a factor, means the factor's *absence* is what does the work. The fill of an arrowhead marks the level of the claim: a hollow head states a causal regularity over event types; a filled head marks the cluster that actually fired in a concrete case — the active cluster.

In the HTML edition the same graphs are interactive: clicking an arc reports the minimally sufficient condition it belongs to, and the derivation steps of a case can be played through as deletions.

2 The paradigm: Manchester Factory Hooters

2.1 Situation

The case owes its canonical role to Mackie (1974, 81–84); Andreas and Günther (2026, sec. 1.3) describe it as adapted from Russell (1921). Every workday at five o'clock the hooters of a Manchester factory sound; a moment later, workers in London stop work and go home. The regular succession is not accidental: knocking-off time at five o'clock sounds the hooters in Manchester and, through the automatic devices that set off the London hooters at five o'clock, sends the Londoners home.

Note

Key. T = knocking-off time (five o'clock) · M = the Manchester factory hooters sound · L = the London workers go home · D = the Manchester hooters are sounded off-schedule (a drill or a fault) · B = an alternative cause of the Londoners' going home (the foreman's early dismissal). $\neg D$ means the hooters were *not* sounded off-schedule. The London hooters, and the devices that set them off at five o'clock, are not a separate factor: they belong to the London-side standing conditions of the causal field and are held fixed.

Mackie himself points out that his INUS analysis counts the sounding of the Manchester hooters as a cause of the Londoners' going home (Mackie 1974, 84; Andreas and Günther 2026, sec. 1.3); Graßhoff and May call the case "a counter example formulated by Mackie himself" (2001, 85). It shows that the INUS analysis certifies as a cause what is not one.

2.2 Causal intuition

The sounding of the Manchester hooters is *not* causally relevant to the London workers' going home. The two events are parallel effects of a common cause. Mackie's own diagnosis appeals to causal priority: preventing the Manchester hooters from sounding does not prevent the Londoners from going home, whereas preventing the London hooters does (Mackie 1974, 85–86; Andreas and Günther 2026, sec. 1.3). The regularity theory does not use this prevention test. In the reconstruction below the London hooters belong to the causal field, not to the factor set, and the verdict is earned by minimal necessity alone.

2.3 Mackie's problem: the INUS analysis certifies the hooters

The INUS idea is that a cause is an *insufficient but non-redundant part of an unnecessary but sufficient condition* (INUS) (Mackie 1965). Consider the conjunction of three conditions: (i) the Manchester hooters sound; (ii) nothing made them sound off-schedule; (iii) the London-side standing conditions, including the devices that set off the London hooters at five o'clock, are in place. This conjunction is sufficient for the Londoners' going home, and no part of it can be dropped without losing sufficiency: it is a cluster, and each of its parts is non-redundant. Thus the sounding of the Manchester hooters passes Mackie's test for being a cause of the Londoners' going home (Mackie 1974, 84).

Andreas and Günther (2026, sec. 1.3) state the structure as two regularities. Written with the letters of the key — their E_1 , E_2 , C_1 are M , L , T , and their auxiliary conditions C_2 (Manchester side) and C_3 (London side) are held fixed in the causal field — they read

$$T + D \leftrightarrow M, \quad T + B \leftrightarrow L.$$

Reading. The Manchester hooters sound exactly when it is knocking-off time or they are sounded off-schedule; the Londoners go home exactly when it is knocking-off time or the foreman dismisses them early (diagram: Figure 3).

Then the cluster $M * \neg D$ is minimally sufficient for L : whenever the hooters sound and were not sounded off-schedule, it is five o'clock, and the Londoners go home; neither conjunct suffices alone. Each conjunct — including M — is an INUS condition of L . Mackie concluded that a pure regularity

theory cannot be the whole story about causation and supplemented it by a notion of causal priority (Mackie 1974, 85–86; Graßhoff and May 2001, 102–3); Baumgartner describes this as abandoning the attempt at a genuine regularity-theoretic analysis of causation (Baumgartner 2008, sec. 7).

2.4 The Graßhoff-May solution: minimal necessity

The solution within the regularity framework is due to Graßhoff and May (2001, 101–4) and May (1999): both call for the minimisation of necessary conditions (Baumgartner 2008, sec. 7). Baumgartner and Graßhoff (2004, 99–103) discuss the case in full (Baumgartner 2008, n. 21).

Mackie minimised only half of his analysis. His test removes idle conjuncts from sufficient conditions, but imposes no test on the *disjunction* of sufficient conditions (the list of alternative causes). The repair is the missing half: a necessary condition must be *minimally necessary* — no disjunct may be deletable without loss of necessity.

Applied to the hooters: the list of putative alternative causes reads

$$(M * \neg D) + T + B \leftrightarrow L.$$

Reading. The Londoners go home exactly when (the hooters sound and were not sounded off-schedule) or it is knocking-off time or the foreman dismisses them early.

Delete the hooters cluster and test what remains: $T + B$ is *still necessary* for L . Whenever the hooters cluster $M * \neg D$ is instantiated, the common cause T is instantiated along with it. The hooters cluster is a free rider, and **minimal necessity** removes it. Since M appears nowhere in the minimal theory of L , the Manchester hooters are not causally relevant to the Londoners' going home.

What the elimination leaves behind is the structure the intuition had all along:

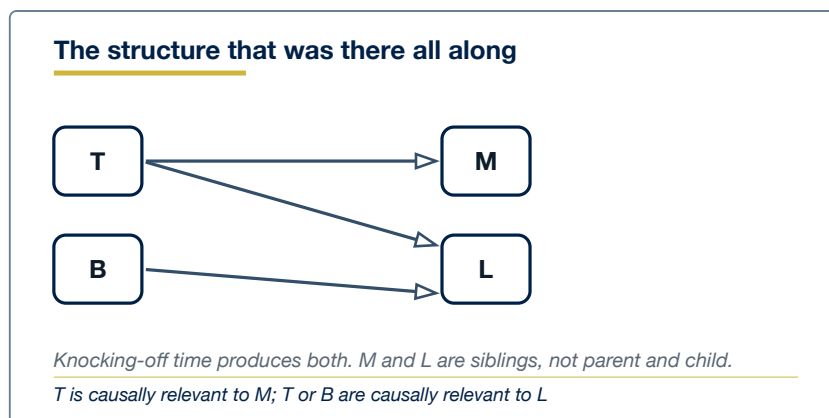


Figure 2 The hooters case after minimal necessity has done its work. Knocking-off time T is causally relevant to the Manchester hooters M and, separately, to the Londoners going home L , for which the foreman's early dismissal B is an alternative cluster; the off-schedule trigger D , the second cluster for M , is left out of the picture. No arrow runs from M to L , and none is needed to account for their perfect correlation: they are two effects of one cause. All arrowheads are hollow, because each arrow states a causal regularity over event types, not what happened on a particular evening. Orientation is a separate matter: L has two clusters, T and B , so the direction of the arrows into L is fixed by the asymmetry of determination, rule (CPLX); the same holds for M once D is counted.

The general lesson is worth stating separately, because the pattern recurs throughout the casebook. Wherever two factors correlate perfectly and one is tempted to call the earlier a cause of the later, minimal necessity asks whether the cluster through the earlier one is *needed* once the common cause is in the theory. If it is not, it goes — and the correlation is explained without a causal link between the correlates.

2.5 Causal analysis: canonical derivation

I Normalisation. Causal field: working days on which the hooter mechanisms function and the London-side standing conditions (Mackie’s C_2 , C_3 , including the devices that set off the London hooters at five o’clock) are in place. $\mathcal{F} = \{T, D, B, M, L\}$.

Symbol	Factor (type)	Token
T	five o’clock (knocking-off time)	present
D	Manchester hooters sounded off-schedule (drill or fault)	absent
B	alternative cause of the Londoners’ going home (foreman’s early dismissal)	absent
M	Manchester factory hooters sound	present
L	London workers go home	present

G Causal graph.

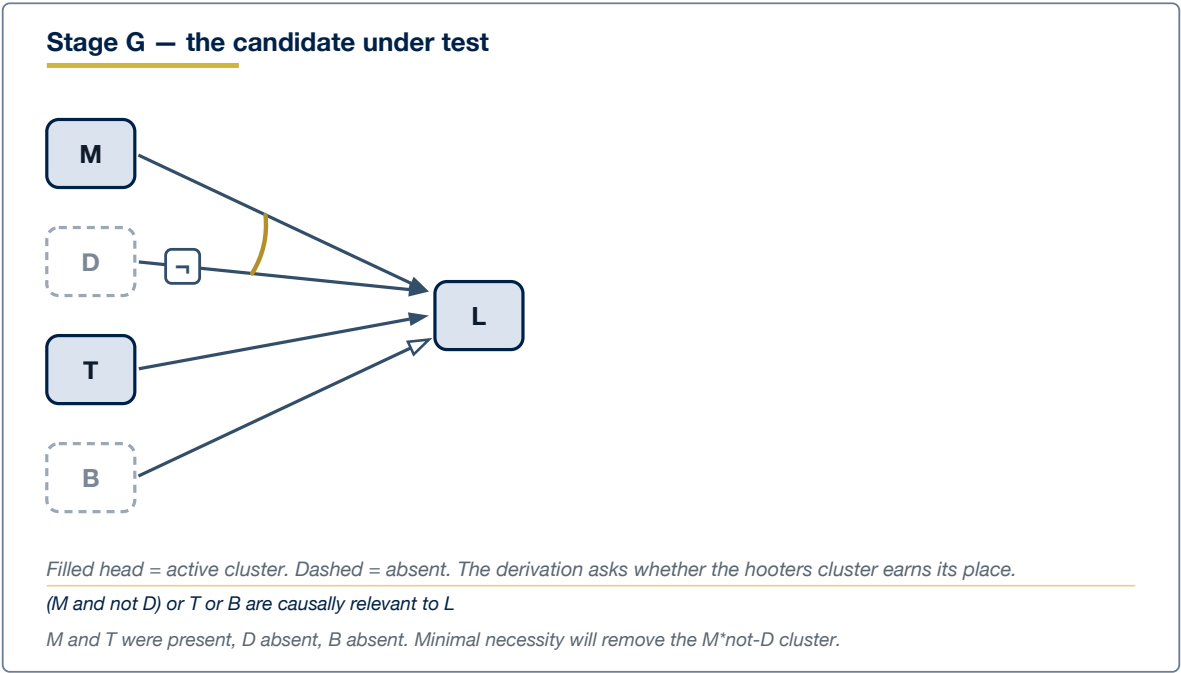


Figure 3 The candidate hypothesis, with this case's token layer marked. The hooters cluster $M * \neg D$ is drawn as one complex cause, on a par with the two alternatives T and B — which is exactly the claim the derivation below tests and rejects. In words: (M and not D) or T or B are causally relevant to L . Filled arrowheads mark the clusters instantiated in the marked row 5, $M * \neg D$ and T ; the hollow head on B marks a cluster that did not fire.

II Coincidence table.

row	T	D	B	M	L
1	0	0	0	0	0
2	0	0	1	0	1
3	0	1	0	1	0
4	0	1	1	1	1
5	1	0	0	1	1
6	1	0	1	1	1
7	1	1	0	1	1
8	1	1	1	1	1

III–IV Model and derivation. MT — Outcome: L .

#	Claim	Licence	In words
1	$T \rightarrow L; B \rightarrow L; M * \neg D \rightarrow L$	(SUF): rows 5–8; rows 2, 4, 6, 8; rows 5, 6	Three conditions are sufficient
2	$T, B, M * \neg D$ are minimally sufficient	(MS): row 3 refutes M alone; row 1 refutes $\neg D$ alone	No conjunct is idle
3	$T + B + M * \neg D \leftrightarrow L$	(NEC): each L -row (2, 4–8) satisfies a disjunct; the rows without L (1, 3) satisfy none	Necessary condition
4	<i>INUS verdict</i> : M is an INUS condition of L	INUS stops at line 3	INUS overgenerates
5	$T + B$ is necessary for L	(NEC): each L -row (2, 4–8) contains T or B	T or B is present whenever L is
6	Eliminate $M * \neg D$: $T + B$ is a necessary	(MN), from lines 3 and 5	Graßhoff–May step

#	Claim	Licence	In words
	proper part of line 3, so line 3 is not <i>minimally</i> necessary		
7	Neither T nor B can be eliminated: row 2 has L without T , row 5 has L without B ; two clusters remain	(MN); (CPLX)(a)	Direction is structurally oriented
8	Minimal theory: $T +$ $B \Rightarrow L$	lines 2, 6, 7: (MS), (MN), (CPLX)	T or B are causally relevant to L

V Verdict. M is not causally relevant to L : knocking-off time or the foreman's early dismissal are causally relevant to the Londoners' going home, and the Manchester hooters occur in no cluster of that minimal theory (Figure 2). The verdict replicates the intuition, and every step is read off the eight rows.

3 Type-Level Benchmark Sprachspiele

3.1 Case 1: Struck match and sung song (Monotony & Minimal Sufficiency)

Situation. A match is struck against a box (S), the head is dry, and oxygen is present. The striker also sings a song (Q). The match ignites (I) (Baumgartner 2008, secs. 3–4).

Intuition. S is causally relevant to I ; Q is not.

Regularity judgement. Minimal sufficiency (MS) prunes Q : S is a conjunct of an MSC for I , whereas Q is idle.

I Normalisation. Causal field: dry match heads, enough oxygen. $\mathcal{F} = \{S, Q, X, I\}$ (X : open flame).

Symbol	Factor (type)	Token
S	match is struck	present
Q	striker sings a song	present
X	match exposed to open flame	absent
I	match ignites	present

G Causal graph.

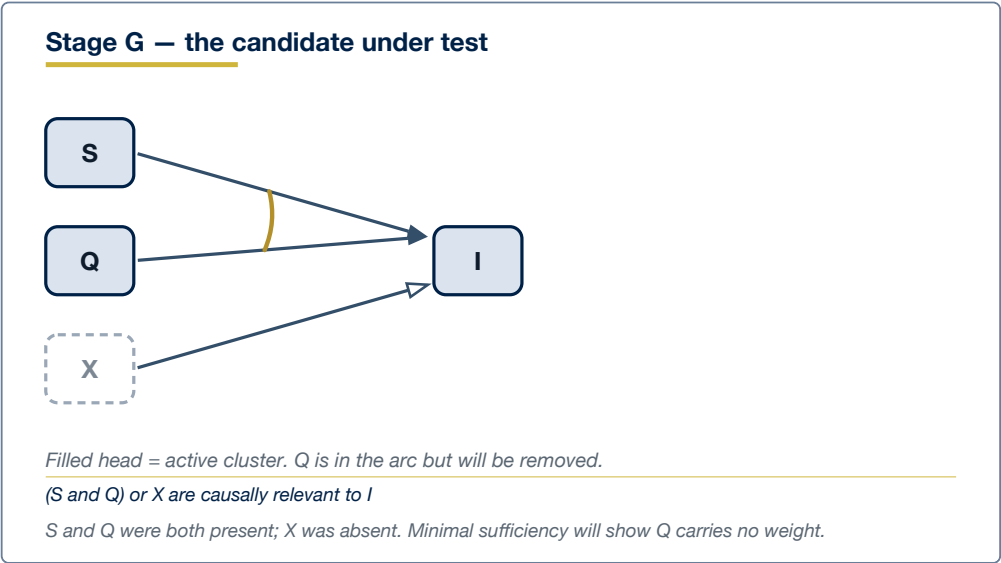


Figure 4 The candidate hypothesis for the match. Striking and singing are tied by an arc, so the candidate treats the song as a required part of the cause. Row 5 of the table — struck, not sung, ignited — is what breaks the arc.

II Coincidence table.

row	S	Q	X	I
1	0	0	0	0
2	0	1	0	0
3	0	0	1	1
4	0	1	1	1
5	1	0	0	1
6	1	1	0	1

III–IV Derivation. MT — Outcome: *I*.

#	Claim	Licence
1	$S * Q \rightarrow I; S \rightarrow I; X \rightarrow I$	(SUF): row 6; rows 5, 6; rows 3, 4
2	<i>Q</i> is idle in $S * Q$; <i>Q</i> alone is not sufficient	(MS): rows 5, 6 (<i>S</i> alone suffices); row 2 refutes <i>Q</i>
3	$S + X$ is necessary for <i>I</i> , and neither disjunct alone is	(NEC), (MN): every <i>I</i> -row (3–6) has <i>S</i> or <i>X</i> ; row 3 lacks <i>S</i> , row 5 lacks <i>X</i>
4	Minimal theory: $S + X \Rightarrow I$	lines 2, 3; (CPLX)(a), two clusters

Reading. Striking or exposure to an open flame are causally relevant to the match's igniting.

V Verdict. *S* is causally relevant to *I*; *Q* is not.

The remaining type-level cases (rows 2–6 of Table 1) vary this pattern. Each tests one condition: equifinality (short circuit and arson), conjunctivity (match and oxygen), chains (switch and lamp), common cause (barometer), orientation (asymmetric structure). They are recorded in the master table with their deciding rule, not restated as empty sections.

4 Token-level actual causation

Type-level membership does not settle which cluster fired in a particular situation. Actual causation asks which instantiated cluster brought about *this* outcome.

4.1 Case 7: Two fires (symmetric overdetermination)

Situation. Two fires, ignited independently by short circuit (*A*) and lightning (*B*), reach a building at the same moment and burn it down (*E*). The structure is Baumgartner's symmetric overdetermination, where two neurons "trigger *E* simultaneously, such that each stimulus would have itself been sufficient for *E* to fire" (Baumgartner 2013, fig. 4c); the fire story is an illustration of it.

Intuition. Both *a* and *b* — the particular short circuit and the particular lightning strike — are actual causes of *e*, the burning of this building.

Regularity judgement. Both clusters fire in this case. Actual causation certifies both clusters as active. The case shows that minimality at type level does not force uniqueness at token level: overdetermination is membership of two instantiated clusters, not a defect of the theory.

4.2 Case 8: Suzie and Billy (late preemption)

Situation. Suzie (*A*) and Billy (*B*) both throw rocks at a bottle; the bottle shatters (*E*). Suzie's rock arrives first and shatters it; Billy's rock passes through empty air. Billy's process is cut off by the effect itself — the pattern Baumgartner analyses as late preemption (Baumgartner 2013, fig. 4b); the rock story is an illustration of it.

Intuition. Suzie's throw caused the shattering; Billy's did not.

Regularity judgement. Suzie's throw belongs to the active cluster. Billy's cluster is preempted: it is not completely instantiated, because his rock never meets an intact bottle. Actual causation tracks the instantiated cluster, not the counterfactual "had Suzie not thrown, Billy would have."

The remaining token-level cases (rows 9–16 of Table 1) vary preemption, switching, prevention, and omission. Each is decided by the rule named in the table.

5 Explanatory 'because'

Andreas and Günther (2019, 2024a, 2024b) extend regularity theory to explanatory 'because' claims. The load-bearing contrast is the tower and its shadow: the tower explains the shadow, the shadow does not explain the tower. The remaining explanatory cases (rows 17–24 of Table 1) — among them the subcause, extended double prevention and the assassin's mentor — are recorded there with

their deciding rules. The Munich–Lund example, with which Andreas and Günther show that the unstrengthened Ramsey test makes ‘because’ trivial (2019, secs. 1, 3.3), is not a row of the table.

6 Master Benchmark Comparison Table & Synthesis

Table 1 Master comparison table of the 24 philosophical benchmark Sprachspiele. “Constructed” marks an illustration of the named structure written for this casebook; it does not occur in the cited literature.

#	Case Name	Source	Key Concept	Intuition	Regularity Verdict	Deciding Rule
1	Struck match	Baumgartner (2008)	Monotony / Idle conjunct	Cause	Replicated	(MS)
2	Short circuit & arson	Constructed (equifinality)	Equifinality	Cause	Replicated	(MN), (CPLX)
3	Match & oxygen	Baumgartner (2008)	Conjunction / Co-factors	Cause	Replicated	(MS), (SUF)
4	Switch & lamp	Constructed (chain)	Causal chain	Cause	Replicated	(CHAIN)
5	Barometer	Graßhoff & May (2001)	Common cause	No cause	Replicated	(PERM), (MN)
6	Asymmetric structure	Baumgartner (2008)	Orientation	Cause	Replicated	(CPLX)
7	Two fires	Baumgartner (2013)	Overdetermination	Cause	Replicated	(AR)
8	Suzie & Billy	Baumgartner (2013)	Late preemption	Suzie cause	Replicated	(AR), (MARK)
9	Major & Sergeant	Baumgartner (2013)	Late preemption	Major cause	Replicated	(AR), (MARK)
10	Merlin & Morgana	Baumgartner & Falk (2023)	Trumping	Merlin cause	Replicated	(CMT)
11	Train switch	Baumgartner (2013)	Switching	No cause	Replicated	(MS)
12	Short circuit & sprinkler	Baumgartner (2013)	Short circuit	No cause	Replicated	(MS)

#	Case Name	Source	Key Concept	Intuition	Regularity Verdict	Deciding Rule
13	Poison & antidote	Baumgartner (2013)	Prevention	Antidote cause	Replicated	(NEG-LIT)
14	Guard asleep	Baumgartner (2013)	Omission	Guard cause	Replicated	(NEG-LIT)
15	Double prevention	Baumgartner (2013)	Double prevention	Cause	Replicated	(DBL-PREV)
16	Preemption by prevention	Baumgartner (2013)	Path preemption	Cause	Replicated	(PATH-PREV)
17	Re-ignition	Andreas & Günther (2019)	Explanatory 'because'	Because	Replicated	(RT-B)
18	Secondary cause	Andreas & Günther (2019)	Background context	Because	Replicated	(RT-B)
19	Negative cause	Andreas & Günther (2019)	Negative explanation	Because	Replicated	(RT-B)
20	Counterfactual because	Andreas & Günther (2019)	Counterfactual match	Because	Replicated	(RT-B)
21	Subcause	Andreas & Günther (2024b)	Entangled causes	Cause	Replicated	(LREL)
22	Extended double prev.	Andreas & Günther (2024b)	Short circuit in DP	<i>c</i> cause, <i>f</i> not	Replicated	(PATH), (MAX)
23	Assassin's mentor	Andreas & Günther (2024a)	Bogus prevention	Antidote not cause	Replicated	(DEV')
24	Tower & shadow	Andreas & Günther (2019)	Explanatory asymmetry	Tower causes	Replicated	(RT-B)

7 Part 3: Qualitative Cases (Volume I Tests)

The following panels apply the definitions of Volume I to coincidence tables. Each is carried through the six-step protocol.

7.1 Case A-01: Warehouse Fire – Is the Short Circuit a Cause?

Case A-01 — Is short circuit (SC) a cause of fire (FI)?

Query. Is SC (short circuit) causally relevant to FI (fire) in the causal field of the nine recorded test situations?

Key. SC = short circuit · OX = oxygen present · FL = flammable liquid present · SP = spark · FI = fire.

Causal field and factor set. $\mathcal{F} = \{\text{SC}, \text{OX}, \text{FL}, \text{SP}\}$, outcome FI; nine test situations. No process order is required: with two equifinal MSCs, structural non-symmetry orients the theory without temporal data (Volume I, novelty 5 of the minimal theory).

Data setup.

Case	SC	OX	FL	SP	FI
1	1	1	0	0	1
2	1	1	1	1	1
3	0	1	1	1	1
4	0	0	0	0	0
5	1	0	0	0	0
6	0	1	0	0	0
7	0	1	1	0	0
8	0	1	0	1	0
9	0	0	1	1	0

Source: constructed test situations for the warehouse-fire structure of Volume I; row 9 is a sealed chamber (flammable liquid and a spark without oxygen). One row per test situation; no replication.

Gate A. Contradiction test: $C_E = 0$, $B_E = 0$. No contradictory configurations. Homogeneity not falsified; proceed to MSC search.

Gate B. Not applicable: this is a qualitative coincidence table, not a replicated experimental study.

MSC search (engine calls).

```
minimal_theory_search(cases, outcomes=["FI"])
# → FL*OX*SP + OX*SC => FI
hg.is_relevant("SC", "FI") # → True [INF-01]
hg.partners("SC", "FI") # → [{OX}] [INF-04]
hg.alternatives("FI") # → [SC*OX, FL*SP*OX] [INF-05]
```

Causal hypothesis.

$$SC * OX + FL * SP * OX \Rightarrow FI$$

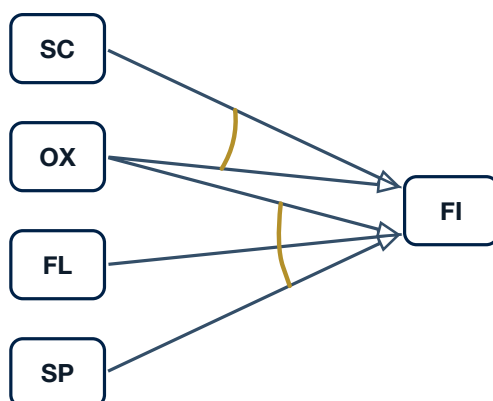
Reading. (SC and OX) or (FL and SP and OX) are causally relevant to FI.

[con = cov = 1.0, $n = 9$, robustness = 1.0, $|\mathcal{A}| = 1$]. SC is a non-redundant literal in the cluster SC * OX: the cluster is sufficient by rows 1 and 2; SC alone is refuted by row 5, OX alone by row 6. OX is its conjunctive partner. The second cluster FL * SP * OX is sufficient by rows 2 and 3, and each of its conjuncts is indispensable: row 7 lacks the spark, row 8 the liquid, row 9 the oxygen. It can produce fire without SC (row 3).

Licensed verdict — YES. SC is causally relevant to FI (INF-01, INF-02, INF-04, INF-05).

What would change the verdict. A test situation SC=1, OX=1, FL=0, SP=0 with FI=0 would contradict row 1, create a contradictory configuration and trigger factor-set expansion (FALL-03). Removing all FL=1, SP=1 cases would remove the equifinality verdict but not SC's relevance. A field with OX=0 throughout would correctly suspend the verdict: no cluster fires, FI is constant, and relevance claims are suspended relative to that field.

The warehouse fire



*Two clusters for fire. Oxygen is a partner in both; neither cluster fires without it.
(SC and OX) or (FL and SP and OX) are causally relevant to FI*

Figure 5 Case A-01 as a graph: the full warehouse-fire theory. It has two arcs and therefore two clusters: short circuit with oxygen, and flammable liquid with a spark and oxygen. The verdict on the query can be read straight off the picture — SC belongs to an arc, so it is causally relevant; OX belongs to both arcs, so

it is a partner in every cluster; and because two clusters lead to FI, the direction of the whole structure is fixed by the asymmetry of determination without any appeal to what happened first.

7.2 Case A-11: Ice Cream and Drowning – Correlation or Causation?

Case A-11 — Is IC (ice cream sales) causally relevant to DR (drowning deaths)?

Query. Is the observed correlation between ice-cream sales and drowning deaths causal? The six configurations below are illustrative.

Causal field and factor set.

Narrow field: $\mathcal{F}_0 = \{\text{IC}, \text{DR}\}$ (IC = ice-cream sales high, DR = drowning deaths high). Initial factor set from naive analysis.

Expanded field: $\mathcal{F}_1 = \{\text{IC}, \text{DR}, \text{HW}, \text{VO}, \text{SW}\}$ where HW = hot weather, VO = beach vendors open, SW = swimming conditions favourable.

Data setup. Six configurations; the HW column belongs to the expanded field, and the VO and SW columns are not reproduced.

row	HW	IC	DR
1	1	1	1
2	1	1	1
3	0	0	0
4	0	0	0
5	1	0	1
6	0	1	0

Provenance: constructed after published aggregate data on seasonal correlations.

Gate A (narrow field). Each IC value occurs with both DR outcomes: IC = 1 with DR in rows 1, 2 and without it in row 6; IC = 0 with DR in row 5 and without it in rows 3, 4. $C_E = 2$, $B_E = 2/6 \approx 0.33$. Homogeneity violated: no minimal theory for DR is recoverable over $\mathcal{F}_0$. Verdict suspended; field expansion mandated.

Gate A (expanded field). Adding HW already removes both contradictions on these rows: $C_E = 0$. Homogeneity not falsified. Proceed to MSC search.

MSC search (expanded field).

```
minimal_theory_search(cases, outcomes=["DR"]) # → HW => DR on the six rows;
                                                #   HW*SW => DR with SW recorded
hg.is_relevant("IC", "DR") # → False    [INF-11]
hg.common_causes("IC", "DR") # → [HW]    [INF-08]
```

Causal hypothesis. On the six rows DR is present exactly when HW is; with SW recorded the theory of drowning is

$$HW * SW \Rightarrow DR.$$

Reading. Hot weather and favourable swimming conditions are causally relevant to drowning deaths. The theory has a single cluster, so its direction comes from the process order — the weather precedes the drownings, rule (CPLX)(b) — not from the table. For ice-cream sales, hot weather with open vendors, $HW * VO$, is one cluster; row 6 (high sales on a cool day) shows that it is not the only one, and the further cluster is not recorded here.

IC is not a literal in any MSC for DR: IC alone is refuted by row 6, and HW suffices without it (rows 1, 2, 5). HW is a common cause of IC and DR.

Licensed verdict — NO (INF-08, INF-10, INF-11). IC is not causally relevant to DR. The association is real but is accounted for by the common cause HW.

Warrants (engine run). INF-03, INF-06, INF-08, INF-09, INF-10, INF-11 (examples/casebook/case_11_just_correlation.py).

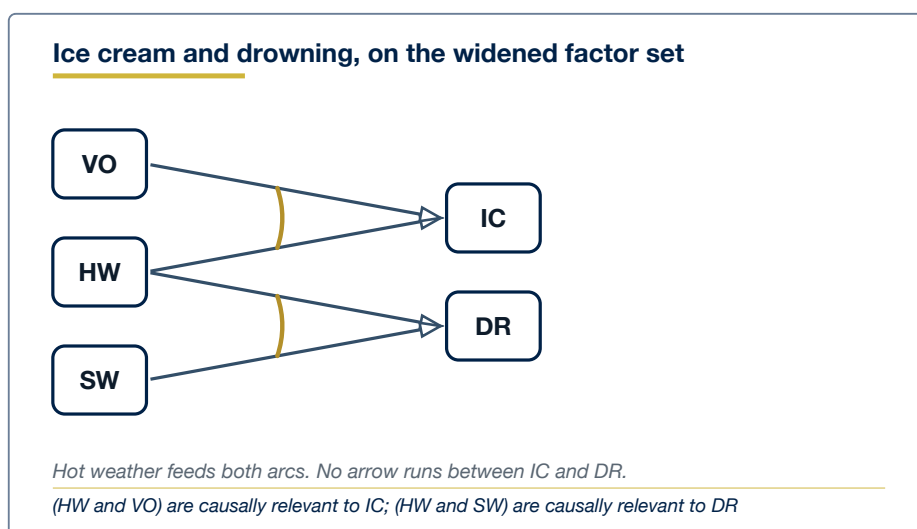


Figure 6 Case A-11 on the widened factor set. Hot weather is a conjunct of the arc into ice-cream sales and of the arc into drownings; there is no arc, and no arrow, running from IC to DR. The correlation is not denied — the picture predicts it, since anything that makes HW true tends to make both outcomes true — but it is produced without a causal link between the two correlates. Compare Figure 2: this is the hooters pattern in epidemiological dress.

FALL codes. FALL-02 (naive correlation without homogeneity check); FALL-03 (contradictory configuration ignored).

What would change the verdict. A difference pair: two cases alike in HW, VO and SW that differ in IC and also differ in DR.

7.3 Case A-FH: Manchester Factory Hooters (Qualitative Panel)

Case A-FH — Do the factory hooters cause workers to wake up?

Query. Are the Manchester factory hooters (FH) causally relevant to workers waking up (WU) in the causal field of the daily production schedule?

Background. A morning variant of the hooters pattern (the canonical case is Section 2): the factory hooters blow at 7:00 am and the workers wake up at 7:00 am, but the hooters do not wake the workers. In the story both follow the schedule; the table decides what that amounts to.

Causal field and factor set. $\mathcal{F} = \{FH, WU, AL, SH\}$. **Key.** FH = factory hooters sound at 7:00 · WU = workers wake up · AL = alarm clock rings · SH = shift change scheduled. Outcome: WU. Process order: SH < FH, SH < AL < WU.

Data setup.

row	SH	FH	AL	WU
1	1	1	1	1
2	0	0	0	0
3	1	1	0	0
4	1	0	1	1
5	0	1	0	0
6	0	1	1	0

Gate A. $C_E = 0$ over $\mathcal{F}$. Homogeneity not falsified.

Gate B. Not applicable (qualitative table).

MSC search.

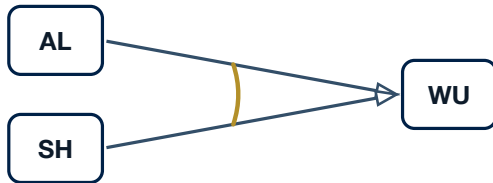
```
minimal_theory_search(cases, outcomes=["WU"]) # → AL*SH => WU
hg.is_relevant("FH", "WU") # → False [INF-11]
```

The alarm with a scheduled shift is sufficient for waking (rows 1 and 4). Neither part suffices alone: row 6 has the alarm without a shift and nobody wakes; row 3 has a shift without the alarm and nobody wakes. $AL * SH$ is therefore minimally sufficient, and it is also necessary: WU occurs only in rows 1 and 4. FH alone is refuted by rows 3, 5 and 6. Two minimally sufficient conditions contain an FH literal, $SH * \neg FH$ and $AL * \neg FH$, each witnessed only by row 4. Both are free riders: $AL * SH$ alone is already necessary for WU, so (MN) removes them.

Causal hypothesis. $AL * SH \Rightarrow WU$.

Reading. (AL and SH) are causally relevant to WU. The theory has a single cluster, so its direction comes from the process order $AL < WU$, rule (CPLX)(b); without a process order the engine reports such a theory with the status UNORIENTABLE. The factory hooters occur in no cluster: they and the waking are both preceded by the scheduled shift, and the table does not show the shift to be a cause of the hooters (row 4 has a scheduled shift without hooters, row 5 hooters without a shift).

Minimal theory of waking: AL and SH are causally relevant to WU



AL = alarm clock rings; SH = shift change scheduled. FH occurs in no cluster: not causally relevant to WU.

$AL \cdot SH \Rightarrow WU$

(AL and SH) are causally relevant to WU

Figure 7 The minimal theory of waking: the alarm clock and the scheduled shift together are sufficient for the workers waking up. In words: (AL and SH) are causally relevant to WU. The factory hooters appear in no cluster.

Licensed verdict — NO. FH is not causally relevant to WU.

Warrants. INF-11 (no cluster with an FH literal survives minimal necessity). The common-cause warrant INF-08 is not licensed by these six rows.

7.4 Engine-Query Crosswalk: Cases 01–20

The full engine-query casebook addresses twenty canonical question-types that arise when reading experimental science.

Table 2 Engine-query crosswalk, Cases 01–20.

Case	Type	Key concept	INF	FALL
01	Qualitative relevance	MSC membership	INF-01–02	—
02	Token attribution	Active MSC	INF-06	—
03	Equifinality	Alternatives	INF-05	—
04	Conjunctivity	Partners	INF-04	—
05	Causal chain	Paths via endogenous	INF-07 (declined)	—
06	Common cause	Joint effects	INF-08	—

Case	Type	Key concept	INF	FALL
07	Direction ambiguity	Asymmetry, csf	INF-03	—
08	Graded evidence	FitVector, robustness	INF-09	—
09	Noisy data	Ambiguity set	—	—
10	Cross-study pooling	E-process	—	—
11	Correlation vs. causation	Naive correlation	INF-11	FALL-02
12	Sample size	Power	—	—
13	Next configuration	EIG	—	—
14	Sequential monitoring	E-process	—	—
15	Post hoc fallacy	No negatives	—	FALL-01
16	Simpson's paradox	Confounder in MSC	INF-03	FALL-04
17	Subgroup effects	HARKing/conjunctivity	INF-04	FALL-09
18	Underpowered study	Absence claim	—	FALL-07
19	Mediation	csf chain	INF-07 (declined)	—
20	Full paper audit	All	All	All

8 Part 4: Quantitative & Experimental Cases (Volume II Tests)

The following panels apply causal hypothesis inference (Volume II) to replicated or published experimental data.

8.1 Case B-ST: Statin Trial – Does STATIN Reduce Cardiovascular Events?

Case B-ST — Does STATIN reduce cardiovascular events in HBP patients?

Query. Is STATIN causally relevant to $\neg$ CVE (event-free survival) in the causal field of the trial population — 400 patients, half of them with a history of high blood pressure?

Causal field and factor set. $\mathcal{F} = \{\text{STATIN}, \text{HBP}, \text{CVE}\}$ — statin therapy, history of high blood pressure, cardiovascular event. Process order: $\text{HBP} < \text{STATIN} < \text{CVE}$ — the history of high blood pressure precedes the therapy, which precedes the event. Polarity: outcome is CVE; causal hypothesis is for the protective direction ($\neg\text{CVE}$).

Coincidence data. 400 patients, 200 per arm. Stratified counts:

	HBP = 1	HBP = 0
STATIN = 1	CVE 5/100	CVE 19/100
STATIN = 0	CVE 30/100	CVE 18/100

Aggregate rates: $p_1 = 24/200 = 0.12$ (STATIN), $p_0 = 48/200 = 0.24$ (control). Each arm holds 100 patients with and 100 without a history of high blood pressure.

Gate B. Randomisation performed; no baseline imbalance in known risk factors. Gate B is satisfied: the difference-test is valid.

Gate A. Within $\mathcal{F} = \{\text{STATIN}, \text{HBP}, \text{CVE}\}$ each of the four configurations of STATIN and HBP occurs with and without CVE: $C_E = 4$, $B_E = 72/400 = 0.18$. Gate A is violated: this factor set is incomplete for sufficiency-reading. That is expected in difference-tests and does not invalidate Gate B.

Risk difference. $\hat{\Delta} = p_1 - p_0 = -0.12$ (protective). Fisher exact $p = 0.003$ (two-sided, 0.0026).

MSC search. The difference-test is carried by the HBP= 1 stratum (5% vs 30%); in the HBP= 0 stratum the rates are 19% and 18%. HBP is therefore the conjunctive partner of STATIN. The engine recovers:

$$\text{STATIN} * \text{HBP} \Rightarrow \neg \text{CVE} \quad [\text{con} = 95/100 = 0.95, \text{cov} = 95/328 \approx 0.29].$$

Reading. Statin therapy and a history of high blood pressure are causally relevant to the absence of a cardiovascular event (diagram: Volume II, the statin figure of the introduction). The coverage of 0.29 records that most of the 328 event-free patients are covered by other clusters for $\neg\text{CVE}$, which this factor set does not name: the absence of an event is equifinal.

Polarity. STATIN is a literal in an MSC for $\neg\text{CVE}$ (protective).

Licensed verdict — YES (protective polarity). Under Gate B, $\hat{\Delta} < 0$ licenses STATIN as a literal of an MSC for $\neg\text{CVE}$, not for CVE: STATIN is a non-redundant literal of the cluster $\text{STATIN} * \text{HBP}$ for the absence of a cardiovascular event.

Warrants. INF-01, INF-04. INF-02 requires consistency 1.0 and is not met ($\text{con} = 0.95$).

8.2 Case B-16: Simpson's Paradox – Kidney Stone Treatments

Case B-16 — Is open surgery (T) causally relevant to treatment success (E) in kidney stone patients?

Query. Is T (open surgery, against percutaneous nephrolithotomy) causally relevant to E (treatment success)? Data: Charig et al. (1986), 700 patients.

Causal field and factor set. $\mathcal{F} = \{T, SEV, E\}$ where SEV = stone severity (large vs. small).

Data setup (stratified 2×2 table).

	Small stones	Large stones	Total
Open surgery	81/87 = 93.1%	192/263 = 73.0%	273/350 = 78.0%
Percutaneous	234/270 = 86.7%	55/80 = 68.8%	289/350 = 82.6%

Open surgery does better in both strata — by 6.4 percentage points for small stones and 4.3 for large ones — but worse overall (−4.6 points), because large stones were assigned to it about 3.3 times as often: 263 of its 350 patients had large stones, against 80 of the 350 percutaneous patients.

Gate B. Non-randomised observational study. SEV co-varies with T assignment — Gate B is violated in the aggregate analysis.

Gate A. Over $\mathcal{F} = \{T, SEV, E\}$: all four configurations of T and SEV carry both outcomes, so $C_E = 4$, and $B_E = (6 + 71 + 36 + 25)/700 = 138/700 = 0.197$. Homogeneity falsified.

Licensed verdict — SUSPENDED. The framework declines to issue a causal verdict on the aggregate data.

FALL codes. FALL-04 (Simpson's paradox, SEVERE): the aggregate risk difference (−4.6 points) has the opposite sign to both strata (+6.4 and +4.3).

Diagnosis. The reversal is compositional confounding. Stone size lowers the success rate under both treatments, and it travels with assignment: large stones went mostly to open surgery. That violates Gate B in the aggregate. It does not show SEV to be a conjunctive partner of T: both strata favour open surgery by similar margins (the two margins differ by 2.2 points), whereas a conjunction $T * SEV$ would show the treatment making its difference in one stratum and not in the other, as HBP does for STATIN in Case B-ST. Within the strata the comparison is closer to Gate B, but in a non-randomised study nothing secures that no further factor travelled with assignment; the verdict therefore stays suspended.

8.3 Case B-18: Underpowered Study – Non-Significance vs. Absence

Case B-18 — Does non-significance establish causal irrelevance?

Query. A trial of $n = 50$ per arm reports 15 events on control against 19 on treatment, $\hat{\Delta} = 0.08$, Fisher exact $p = 0.53$. Does this establish that T is not causally relevant to E?

Causal field and factor set. $\mathcal{F} = \{T, E\}$. Outcome E: adverse event.

Data setup. $n = 50$ per arm. Control rate 30%, treatment rate 38%.

Gate B. Randomisation performed; Gate B secured by design.

Power analysis. Achieved power is 0.13. Detecting a difference of 0.08 from a control rate of 0.30 at 80% power and two-sided $\alpha = 0.05$ requires $n = 550$ per arm by the two-proportion formula of Volume II (section “Power from the identity”). The study is severely underpowered.

Licensed verdict — UNCERTAIN / INCONCLUSIVE. Non-significance in an underpowered study does not license a claim of absence.

FALL codes. FALL-07 (underpowered absence claim); FALL-08 (absence of evidence taken as evidence of absence).

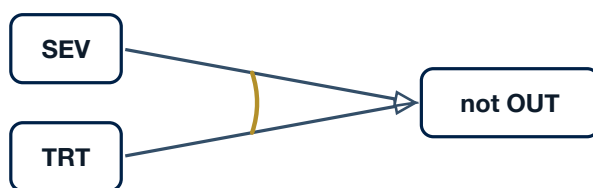
8.4 Case B-20: Full Paper Audit – CONSORT-Compliant RCT

Case B-20 — Full CONSORT-Compliant Paper Audit

Query. Execute a complete 6-stage paper audit on a published CONSORT-compliant clinical trial paper.

Execution steps:

1. **Upload & Storage:** PDF ingested, stored at `papers/paper_b20.pdf`, SHA-256 hash verified.
2. **Metadata Capture:** Title, authors, DOI, trial registration (ClinicalTrials.gov), sample size ($N = 450$), arms, primary outcome recorded.
3. **PDAI Generation:** PDAI-compliant Markdown structure generated with explicit headers and YAML front-matter.
4. **Preprocessing:** Factors extracted ($\mathcal{F} = \{\text{TRT}, \text{AGE}, \text{SEV}, \text{OUT}\}$), binarised, process order assigned ($\text{AGE}, \text{SEV} < \text{TRT} < \text{OUT}$).
5. **Causal Analysis:**
 - *Gate B:* Randomisation and double-blinding secure Gate B.
 - *Gate A:* $C_E > 0$ (expected in clinical trial).
 - *Risk Difference:* $\hat{\Delta} = -0.15$ ($p < 0.001$).
 - *MSC Recovery:* $\text{TRT} * \text{SEV} \Rightarrow \neg \text{OUT}$ (con = 0.94). Key: TRT = treatment; SEV = high severity; OUT = adverse outcome; `~OUT` = absence of the adverse outcome.
6. **Report Generation:** Complete audit report rendered with the warrants INF-01 and INF-04 and zero FALL violations; INF-02 requires consistency 1.0 and is not met at con = 0.94.

CONSORT case: treatment with severity

TRT = treatment; SEV = severity; OUT = adverse outcome. The absence of OUT is the regularity's target.

$TRT * SEV \Rightarrow \sim OUT$

(TRT and SEV) are causally relevant to not OUT

Figure 8 Minimal theory recovered by the audit: treatment combined with high-severity status produces the absence of the adverse outcome. In words: TRT and SEV are causally relevant to the absence of OUT.

8.5 Case B-LK: LK-99 Ambient Superconductivity Audit

Case B-LK — LK-99 Ambient Superconductivity Claim Audit

Query. Does the material LK-99 exhibit ambient-pressure superconductivity (zero resistance, $ZERO_R = 1$)?

Background. Sukbae Lee et al. (2023) claimed zero resistance and levitation in $Pb_{10-x}Cu_x(PO_4)_6O$.

Factor set & Data setup. $\mathcal{F} = \{LK99, CU_SUB, CU2S, ZERO_R, DROP\}$. **Key.** LK99 = sample of the LK-99 compound · CU_SUB = copper substituted on lead sites · CU2S = Cu_2S impurity present · ZERO_R = zero resistance · DROP = sharp drop in resistivity near 377 K.

- Original claim: $LK99 * CU_SUB \Rightarrow ZERO_R$ — the compound with copper substitution is sufficient for zero resistance.
- Replication base: pure single crystals of LK-99 without Cu_2S , grown at the Max Planck Institute for Solid State Research, Stuttgart (Puphal et al., 2023); the identification of the Cu_2S phase transition as the source of the resistivity drop (Jain, University of Illinois Urbana-Champaign, 2023), and tests of samples with different Cu_2S content by a team at the Chinese Academy of Sciences (Zhu et al., 2023).

Gate A Analysis. Pure LK-99 single crystals ($CU2S = 0$) are insulators. Samples with Cu_2S impurities show a sharp resistivity drop near 377 K ($104^\circ C$), the temperature of the known structural phase transition of Cu_2S . No sample shows zero resistance.

Over the claim's own factors $\{LK99, CU_SUB\}$, the configuration $LK99 = 1, CU_SUB = 1$ occurs with the resistivity drop (impure samples) and without it (pure crystals): a contradictory configuration, $C_E > 0$. Adding CU2S to the factor set resolves it: the drop occurs exactly when $CU2S = 1$.

Licensed verdict — FALSIFIED. The claim $LK99 * CU_SUB \Rightarrow ZERO_R$ is refuted by the replication record: the compound is present without zero resistance. The resistivity drop is accounted for by the Cu_2S phase transition, $CU2S \Rightarrow DROP$ (in words: the Cu_2S impurity is causally relevant to the drop; one cluster, oriented by the process order of heating through the transition).

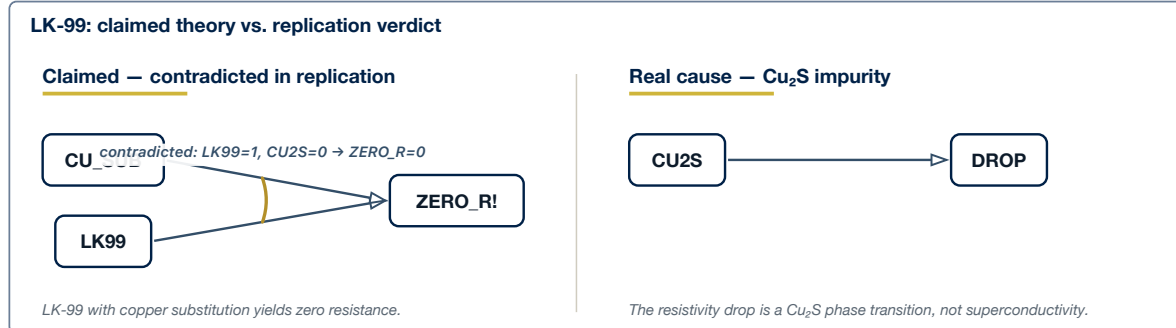


Figure 9 Claimed theory vs. replication verdict. Left: the original hypothesis is contradicted by replication data (pure samples without the Cu_2S impurity show no zero resistance). Right: the confirmed minimal theory assigns DROP to the Cu_2S phase transition alone.

FALL codes. FALL-03 (confounding: contradictory configuration in the replication record), FALL-02 (naive correlation: the drop was attributed to LK99 without a sufficient condition that survives the replication record).

8.6 Case B-OP: OPERA Faster-Than-Light Neutrino Audit

Case B-OP — OPERA Faster-Than-Light Neutrino Speed Audit

Query. Do muon neutrinos travel faster than light ($v > c$)?

Background. The OPERA Collaboration (2011) reported that neutrinos travelling from CERN to Gran Sasso arrived 60.7 ns earlier than light would.

Factor set & Data setup. $\mathcal{F} = \{NEUTRINO, FIBER_LOOSE, CLOCK_CAL, SUPERLUMINAL\}$.

Key. NEUTRINO = a CNGS muon-neutrino event · FIBER_LOOSE = the optical fibre to the master clock improperly connected · CLOCK_CAL = the master-clock oscillator off its calibration · SUPERLUMINAL = an early arrival measured.

Gate A & Diagnostic Analysis. In 2012 OPERA identified two instrumental effects of opposite sign:

1. An improperly connected optical fibre bringing the GPS time signal to the OPERA master clock, which added 73.2 ns to the timing chain and made the neutrinos appear to arrive early.
2. A mis-calibrated 10 MHz oscillator of the master clock, running slightly above its specified frequency, which lengthened the measured time of flight.

With both corrections the published result is an early arrival of $\delta t = (6.5 \pm 7.4 \text{ (stat.) } ^{+8.3}_{-8.0} \text{ (sys.)})$ ns, consistent with $v = c$.

Licensed verdict — FALSIFIED / SYSTEMATIC ARTEFACT. Neutrinos do not exceed light speed. The 60.7 ns anomaly was an artefact of the timing chain, chiefly the improperly connected fibre.

FALL codes. The fallacy registry (FALL-01–10) has no inhibitor for a systematic measurement artefact; the case marks that gap.

9 Part 5: Survey of Experimental Designs and ROADMAP

9.1 Theoretical Survey of Standard Experimental Designs

Every standard experimental design is an instance of the firing-rate identity under a specific Gate-B assignment and sampling mechanism. A randomised controlled trial (RCT) assigns T independently of the latents; a cohort stratifies; a case-control samples on the outcome. The table states what each design identifies and which fallacy it typically risks, with the registry inhibitor where the registry has one.

Table 3 Theoretical survey of six standard experimental designs.

Design Type	Table Structure	Gate B Mechanism	Identifiable Causal Object	Primary risk (registry inhibitor)
RCT	Replicated 2×2 or $2 \times k$ table	Random assignment of T	MSC-membership of T ; risk difference Δ	HARKing, subgroup claims (FALL-09)
Factorial	2^k cross-tabulation	Joint random assignment	Interaction sign I : conjunctive vs. disjunctive	Ignoring the interaction sign (no registry inhibitor)
Cohort	Prospective follow-up table	Stratification & Cornfield bound	Conditional MSC-membership	Unmeasured confounder (FALL-03); Simpson’s reversal (FALL-04)
Case-Control	Retrospective outcome-sampled table	Rare-disease assumption	Risk difference sign (polarity) via OR	Claiming a magnitude from the odds ratio (no registry inhibitor)
Sequential	Longitudinal stream of n trials	Supermartingale e-process	Anytime-valid evidence that a	Optional stopping (no

Design Type	Table Structure	Gate B Mechanism	Identifiable Causal Object	Primary risk (registry inhibitor)
			cluster’s success rate exceeds the base rate	registry inhibitor)
Meta-Analysis	Multi-study pooled table	Random-effects τ^2	Heterogeneous MSC-partners across fields	Pooling inhomogeneous fields (no registry inhibitor)

9.2 The ROADMAP Survey Across Domains

9.2.1 Tranche A: Physics & Materials Discovery

Table 4 ROADMAP Tranche A — Physics & Materials Discovery.

ID	Domain / Case	Factor Set $\mathcal{F}$	Method / Design	Status	Verdict
A-01	Warehouse Fire	SC, OX, FL, SP	Qualitative Coincidence	Complete	SC relevant; SC * OX + FL * SP * OX $\Rightarrow$ FI
B-LK	LK-99 Superconductivity	LK99, CU_SUB, CU2S, ZERO_R, DROP	Physical Replication Audit	Complete	Falsified (Cu ₂ S artefact)
B-OP	OPERA Neutrino Speed	NEUTRINO, FIBER_LOOSE, CLOCK_CAL, SUPERLUMINAL	Systematic Error Audit	Complete	Falsified (timing-chain artefact)
A-02	Room-Temp Hydride	PRESS, TEMP, STRUCT, SC	High-Pressure Diamond Anvil	Ingest Pending	Target: H ₃ S / LaH ₁₀ phase boundary
A-03	Metallic Hydrogen	PRESS, SPECT, TRANS, MH	Optical Absorption	Ingest Pending	Target: Phase transition boundary

9.2.2 Tranche B: Clinical Trials & Epidemiological Cohorts

Table 5 ROADMAP Tranche B — Clinical Trials & Epidemiological Cohorts.

ID	Domain / Case	Factor Set $\mathcal{F}$	Method / Design	Status	Verdict
B-ST	Statin Trial	STATIN, HBP CVE	Randomised Trial (RCT)	Complete	Protective: STATIN * HBP $\Rightarrow$ $\neg$ CVE
B-16	Simpson's Paradox	T, SEV, E	Stratified Cohort	Complete	Suspended (compositional confounding)
B-18	Underpowered Study	T, E	Low-power RCT	Complete	Inconclusive (FALL-07)
B-20	CONSORT Paper Audit	TRT, AGE, SEV, OUT	Full Paperflow Audit	Complete	Automated 6- stage report
B-01	SGLT2 Inhibitor Trial	SGLT2, CKD, T2D, RENAL	Multicentre RCT	Ingest Pending	Renal protection MSC
B-02	mRNA Vaccine Efficacy	VAC, AGE, VAR, INF	Phase III RCT + Cohort	Ingest Pending	Waning & variant interaction

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